

Linux Security Monitoring & Incident Detection using Splunk

A hands-on SOC-focused project demonstrating Linux authentication monitoring, privilege escalation detection, alerting, and correlation using Splunk Cloud.

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Target Role: SOC Analyst / Jr. Security Analyst

Tools Used: **Splunk Cloud, Linux**

1. Introduction

Security Operations Centers (SOCs) continuously monitor system logs to detect suspicious activities and potential security incidents. Linux systems generate critical authentication and privilege-related logs that are often targeted during unauthorized access attempts.

This project focuses on building a Linux security monitoring solution using Splunk Cloud to detect failed login attempts, sudo abuse, user creation events, and correlated attack behavior. The objective is to simulate real SOC workflows such as log analysis, alert creation, dashboard visualization, and incident correlation using realistic Linux attack scenarios.

2. Environment and Tools Used

The project was implemented in a controlled lab environment to simulate realistic Linux security monitoring scenarios.

- Operating System: Ubuntu (local test machine)
- Log Sources: Linux authentication and system logs
- SIEM Platform: Splunk Cloud (Free Trial)
- Data Ingestion Method: Splunk Universal Forwarder / Syslog ingestion
- Dashboard Type: Splunk Dashboard Studio
- Alerting: Scheduled and manually tested alerts

This environment was designed to replicate common SOC monitoring conditions for Linux-based systems while maintaining simplicity and control for testing detection logic.

3. Log Sources and Data Collection

Linux systems generate critical logs related to authentication, authorization, and system activity. These logs are essential for detecting unauthorized access and privilege escalation attempts.

The following log types were ingested into Splunk for analysis:

- Authentication logs capturing failed and successful login attempts
- Sudo logs capturing both failed and successful privilege escalation
- User management events such as new user creation
- Command execution logs related to administrative actions

These logs were continuously indexed in Splunk, allowing real-time searching, visualization, and alerting based on suspicious behavior.

4. Detection Use Cases

This section describes the security use cases implemented in Splunk to detect suspicious authentication and privilege-related activity on a Linux system. Each use case represents a common SOC monitoring scenario and is supported by SPL queries, dashboard visualizations, and alerts.

4.1 Failed Login Detection

Objective:

To detect repeated failed authentication attempts that may indicate brute-force attacks, credential guessing, or unauthorized access attempts on a Linux system.

Detection Logic:

Linux authentication logs were analyzed to identify failed login events. These events were aggregated by user and time to observe patterns of repeated failures.

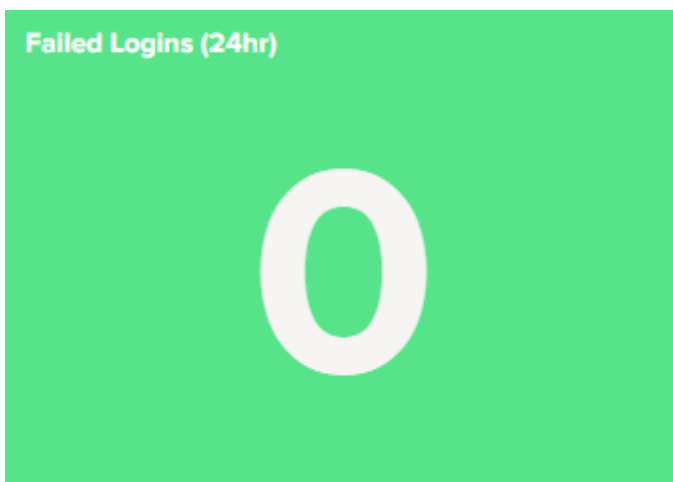
Sample SPL Query:

```
index=* sourcetype=linux_secure  
| search "Failed password" OR "authentication failure"  
| stats count
```

Dashboard Visualization:

This detection is visualized using a KPI and count of failed login attempts over time.

Screenshot:



4.2 Failed Sudo Attempts

Objective:

To detect failed privilege escalation attempts where a user attempts to execute commands using sudo without proper authorization.

Detection Logic:

Sudo log entries were filtered for authentication failures, indicating incorrect password attempts during sudo execution.

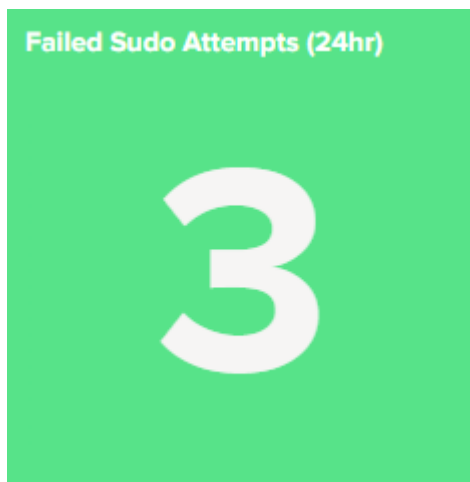
Sample SPL Query:

```
index=*main | search "incorrect password" | stats count
```

Dashboard Visualization:

A KPI and statistical panel were used to highlight users generating failed sudo attempts.

Screenshot:



4.3 Successful Sudo Activity

Objective:

To monitor successful privilege escalation events that could indicate legitimate administrative actions or potential misuse of elevated privileges.

Detection Logic:

Successful sudo events were tracked to identify which users gained root-level access and how frequently.

Sample SPL Query:

```
index=* sudo  
| search "session opened for user root"  
| stats count
```

Dashboard Visualization:

This detection is represented using a statistical panel showing users with successful sudo activity.

Screenshot:



4.4 New User Creation Monitoring

Objective:

To detect the creation of new user accounts, which may indicate legitimate administrative actions or unauthorized persistence mechanisms used by attackers.

Detection Logic:

User management logs were monitored for account creation events, and alerts were configured to trigger immediately upon detection.

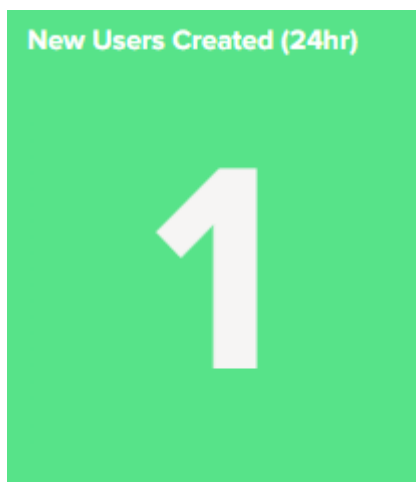
Sample SPL Query:

```
index=* useradd  
| stats count
```

Dashboard Visualization:

This use case is shown as a KPI and event count panel to display the total volume of new user creation attempts.

Screenshot:



5. Alerts Configuration and Testing

Alerts were configured in Splunk to automatically detect and notify suspicious activity based on predefined thresholds. Each alert was tested using controlled, simulated attack activity to ensure that the detection logic functioned correctly.

5.1 Failed Sudo Attempts Alert

Purpose:

To alert when repeated failed sudo attempts occur, which may indicate unauthorized privilege escalation attempts.

Trigger Condition:

Alert triggers when one or more failed sudo authentication events are detected within the defined monitoring window.

Alert Testing:

The alert was tested by intentionally entering incorrect passwords during sudo execution to confirm that the alert fired as expected.

Screenshot:

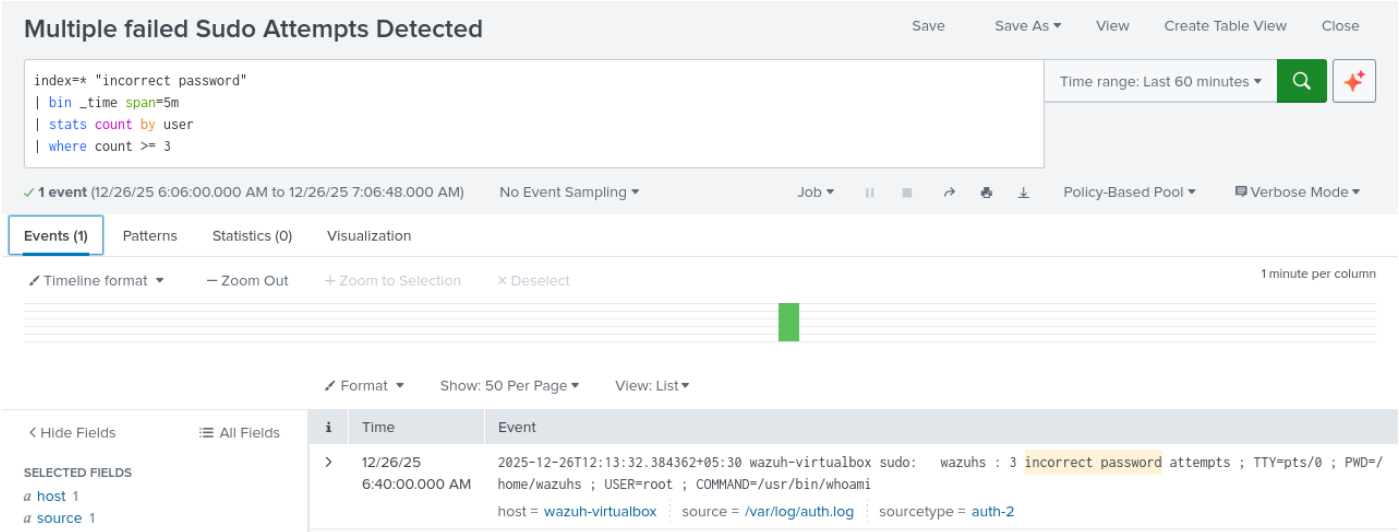


Fig 5.1: Failed Sudo Attempts

5.2 New User Creation Alert

Purpose:

To immediately detect the creation of new user accounts, as this activity may indicate persistence mechanisms or unauthorized access.

Trigger Condition:

Alert triggers upon detection of user creation events in Linux system logs.

Alert Testing:

The alert was tested by manually creating a new user on the system and verifying that the alert was generated.

Screenshot:

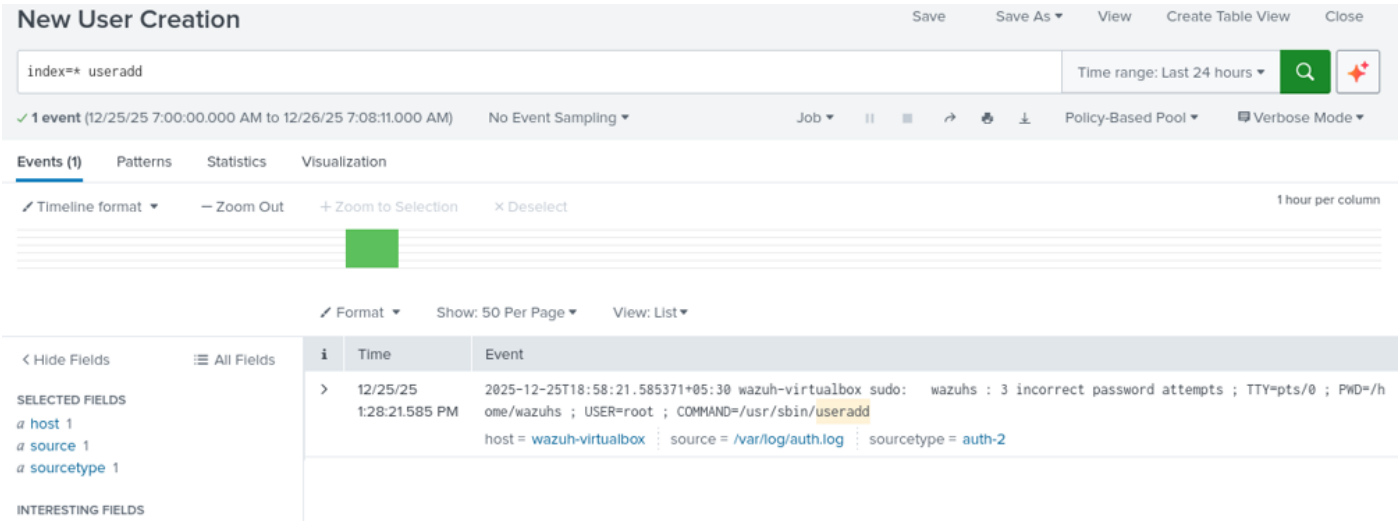


Fig 5.2: New User Creation Alert

5.3 Correlated Privilege Escalation Alert

Purpose:

To identify suspicious behavior by correlating multiple attack stages on a single host, reducing false positives caused by isolated events.

Trigger Condition:

Alert triggers when two or more distinct attack stages (e.g., failed login followed by failed or successful sudo) occur on the same host within a defined time window.

Alert Testing:

The alert was validated by simulating multiple attack stages and confirming that the correlation alert fired correctly.

Screenshot:

Suspicious Privilege Escalation Sequence

SaveSave AsViewCreate Table ViewClose

Suspicious Privilege Escalation Sequence

Time range: Last 1 hour

```
index=*
("incorrect password" OR "session opened for user root")
| eval attack_stage=case(
  like(_raw,"%incorrect password%"), "Failed Sudo Attempt",
  like(_raw,"%session opened for user root%"), "Successful Sudo"
)
| stats
  values(attack_stage) as attack_stages
  min(_time) as first_seen
  max(_time) as last_seen
  count as event_count
  by host
| where mvcount(attack_stages)=2
| eval duration=last_seen-first_seen
| eval first_seen=strftime(first_seen,"%Y-%m-%d %H:%M:%S"),
  last_seen=strftime(last_seen,"%Y-%m-%d %H:%M:%S")
```

70 events (12/26/25 6:09:00.000 AM to 12/26/25 7:09:00.000 AM)No Event SamplingJob|||Policy-Based PoolVerbose Mode

Events (70)PatternsStatistics (1)Visualization

Show: 20 Per PageFormatPreview: On

host	attack_stages	first_seen	last_seen	event_count	duration
wazuh-virtualbox	Failed Sudo Attempt Successful Sudo	2025-12-26 06:09:01	2025-12-26 07:08:01	70	3540.441405

6. Dashboard Design and Visualization

A centralized security dashboard was designed to provide SOC-style visibility into Linux authentication and privilege-related activity. The dashboard enables quick triage, trend analysis, and investigation without requiring analysts to manually search logs continuously.

The dashboard is structured into the following logical sections:

• **KPI Overview:**

Displays high-level security indicators such as failed logins, failed sudo attempts, successful sudo activity, and new user creation.

• **Activity Timeline:**

Time-based visualizations show trends in authentication and sudo activity, helping identify spikes or abnormal behavior.

• **Investigation Panels:**

Statistical tables provide detailed context such as affected users and hosts, supporting deeper investigation when alerts are triggered.

• **Correlated Incident View:**

A dedicated panel highlights hosts exhibiting multiple attack stages, allowing analysts to focus on high-risk incidents rather than isolated events.

Screenshot:

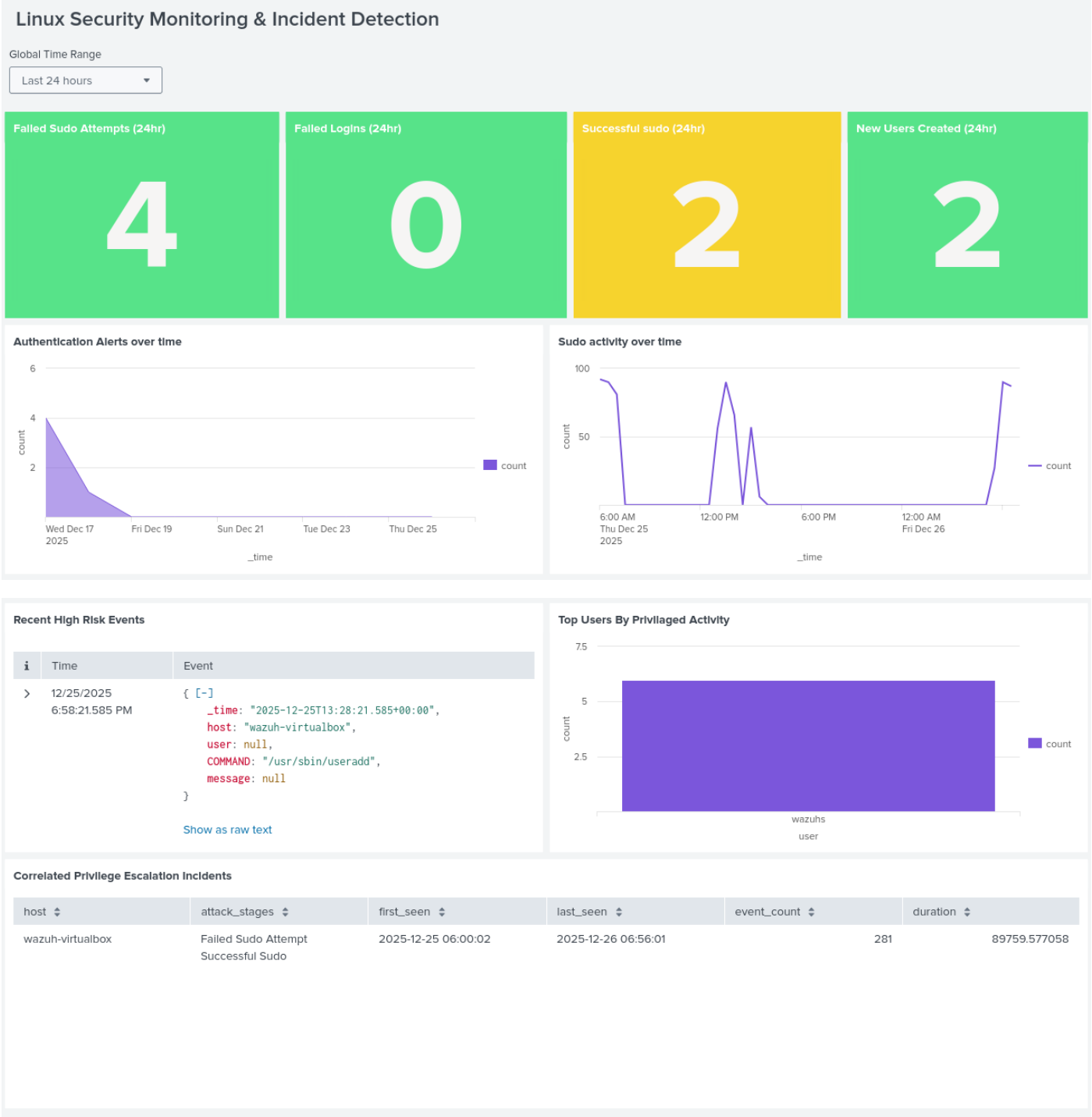


fig 6.1: Linux Security Monitoring & Incident Detection dashboard

7. Correlation Logic for Privilege Escalation Detection

In real-world SOC environments, single security events often generate noise and do not necessarily indicate malicious activity. To reduce false positives, this project implements correlation logic to identify meaningful attack patterns.

Multiple security events such as failed login attempts, failed sudo attempts, and successful sudo activity were categorized as individual attack stages. These stages were then correlated based on the host and time window to identify potential privilege escalation attempts. Only hosts exhibiting two or more distinct attack stages were flagged as high-risk, allowing analysts to focus on actionable incidents rather than isolated events.

Screenshot:

New Search Save As ▾ Create Table View Close

Time range: Last 24 hours 🔍 🌟

```
index=*
("incorrect password" OR "session opened for user root")
| eval attack_stage=case(
  like(_raw,"%incorrect password%"), "Failed Sudo Attempt",
  like(_raw,"%session opened for user root%"), "Successful Sudo"
)
| stats
  values(attack_stage) as attack_stages
  min(_time) as first_seen
  max(_time) as last_seen
  count as event_count
by host
| where mvcount(attack_stages)=2
| eval duration=last_seen-first_seen
| eval first_seen=strftime(first_seen,"%Y-%m-%d %H:%M:%S"),
  last_seen=strftime(last_seen,"%Y-%m-%d %H:%M:%S")
```

✓ 231 events (12/25/25 7:00:00.000 AM to 12/26/25 7:13:04.000 AM) No Event Sampling ▾ Job ▾ || ■ ↶ ↷ ⬇ Policy-Based Pool ▾ Verbose Mode ▾

Events (231) Patterns **Statistics (1)** Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On

host	attack_stages	first_seen	last_seen	event_count	duration
wazuh-virtualbox	Failed Sudo Attempt Successful Sudo	2025-12-25 07:00:01	2025-12-26 07:13:01	231	87179.973594

Fig 7.1: Correlation Search

8. Limitations of the Project

- The project was implemented in a single-host lab environment.
- Log sources were limited to Linux authentication and system logs.
- Attack activity was simulated for testing purposes.
- No automated response (SOAR) was implemented.
- Thresholds were chosen for demonstration and may require tuning in production environments.

9. Conclusion

This project demonstrates the practical application of SIEM concepts using Splunk for Linux security monitoring. By implementing dashboards, alerts, and correlation logic, the project reflects core SOC workflows such as detection validation, alerting, and incident identification.

The experience gained through this project has strengthened my practical understanding of log analysis, security monitoring, and SOC operations.